

College Education, Enrollment Location, and the Geographical Mobility of Young Adults*

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Abstract

Does higher education universally raise geographic mobility of young adults? We revisit this question by separating the effect of earning a degree from the effect of relocating to attend college. Exploiting China's large-scale higher education expansion as a natural experiment, we isolate general human capital (the degree) from place-specific exposure (the enrollment location). We find a sharp contrast: out-of-province enrollment substantially increases subsequent moves, whereas holding a college degree, by itself, does not. We reconcile this pattern by showing that each margin shifts perceived benefits and costs of moving through job opportunities, institutional frictions, and social norms. These results refine our understanding of human capital sorting and inform policies aimed at managing regional brain drain.

Keywords: College Education, Enrollment Location, Migration, Higher Education Expansion
JEL Classification Codes: I23, J61, O15, R23

*This paper has benefited from comments by Jorge De la Roca, Jingting Fan, Min Fang, Suqin Ge, Yilin Hou, Zibin Huang, Shumin Carl Lin, Ming Lu, Amanda Ross, Justin L. Tobias, Le Wang, John Winters, Chunbin Xing, Yu Yang, Junsen Zhang, Xiaohan Zhong, and seminar and conference participants at the SEA Annual Meeting, AEA Annual Meeting, SRSA Annual Meeting, NARSC Annual Meeting, CES Conferences, the Annual Workshop on Economics of Education in China 2025, and Renmin University of China. Shu Cai acknowledges financial support from the China Natural Science Foundation (Project No. 72173056) and the Fundamental Research Funds for the Central Universities (Project No. 23JNQMX29). All remaining errors are our own.

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1 Introduction

The migration of skilled young adults is central to regional labor market integration and economic development, shaping the spatial distribution of talent and aggregate productivity (e.g., [Au and Henderson, 2006](#); [Li, Whalley, and Xing, 2014](#); [Che and Zhang, 2018](#); [Bryan and Morten, 2019](#); [Tombe and Zhu, 2019](#); [Howard, 2020](#); [Diamond and Gaubert, 2022](#); [Fabre, 2023](#)). Regions that attract college graduates gain in productivity, innovation, and knowledge diffusion (e.g., [Glaeser, Scheinkman, and Shleifer, 1995](#); [Moretti, 2004, 2011](#); [Bosetti, Cattaneo, and Verdolini, 2015](#); [Ma, 2024](#)), while those with net outflows risk brain drain ([Beine, Docquier, and Rapoport, 2001, 2008](#)).

Policy responses—such as national enrollment quotas, investments in public institutions, and talent incentives—hinge on a key distinction: the effect of education itself versus the effect of living in the place of enrollment. Disentangling the education effect *per se* (a “general human capital” channel) from the enrollment location effect *per se* (a “location experience” channel) is essential for designing interventions that improve the allocation of talent ([Groen, 2011](#)).

This distinction bears on a central puzzle in spatial macroeconomics: why large regional income differences persist and why migration responses to shocks are often slow. Recent work highlights “local ties”—non-monetary attachments that keep workers in their home regions despite better opportunities elsewhere ([Zabek, 2024](#)). Yet the origins of these ties, and how they are shaped by foundational human capital investments, remain unclear.

Despite the importance of this distinction, empirical research has rarely isolated the separate effects of education versus enrollment location, typically estimating only their combined influence ([Machin, Pelkonen, and Savanes, 2012](#); [Malamud and Wozniak, 2012](#); [Kennan, 2015](#); [Winters, 2020](#); [Ding, 2021](#)). We address this gap by providing the first rigorous, separate causal estimates for each channel, leveraging the unique setting of China. Our paper thus provides a micro-foundation for spatial macroeconomics by investigating whether education policy itself shapes the local attachments that drive the macroeconomic outcomes.

This endeavor faces two key empirical challenges. First, the scarcity of longitudinal residential data in developing economies makes it difficult to observe key life-stage locations ([Kong, Meng, and Zhang, 2009](#); [Giles et al., 2012](#)). We overcome this by developing a novel method to infer college enrollment locations from retrospective questions in China’s population censuses. Second,

education and migration choices are not random, but are correlated with unobserved factors such as innate ability, hometown ties, or local social welfare programs that differentially reward educational attainment, creating a significant risk of spurious correlation (Notowidigdo, 2020).

To address endogeneity, we implement an instrumental variables (IV) strategy that exploits China’s late-1990s higher-education expansion as a natural experiment. A key feature of this reform is that it generated two distinct, plausibly exogenous shocks through the provincial quota system: one to the supply of seats at local universities, and another to the supply of seats allocated to local students at out-of-province universities. These shocks create sharp variation in college access across provinces and cohorts and provide a compelling setting to study the determinants of human capital allocation.¹ Specifically, leveraging this dual variation, we instrument for two endogenous choices—college attendance and out-of-province enrollment—enabling us to disentangle the causal effect of education *per se* from the effect of enrollment location on post-graduation migration.

We implement a two-step IV approach suited to models in which endogenous regressors arise from nonlinear processes—here, binary educational choices (Wooldridge, 2010). In the first step, we use the college seat supply shocks to predict individual propensities for local and out-of-province enrollment. In the second step, these predicted probabilities are used as instruments for actual enrollment decisions to estimate their causal impacts on migration.

Identification rests on the exclusion restriction, which requires that regional seat supply shocks affect later-life migration only through decisions about college attendance and enrollment location. This identifying assumption is strongly supported by the large, centrally planned, and abrupt nature of the expansion at the macrogeographic level, which limits correlation with unobserved individual location preferences. Furthermore, as we show later in the paper, empirical tests corroborate the exogeneity of the shocks by revealing no evidence of differential pre-trends prior to the policy’s implementation.

Our results reveal two sharply contrasting effects. First, we find a powerful “location experience” effect that attending college outside one’s home province substantially increases the likelihood of post-graduation migration. A one-percentage-point increase in the probability of out-of-province enrollment raises the postcollege migration probability by approximately three

¹China’s administrative divisions are organized into provinces, prefectures, counties, townships, and villages, with provinces being the focus of this study due to data availability. For comparison, a province in China can be loosely equated to a state in the U.S.

percentage points, net of the education effect. This is driven by how the experience of living elsewhere for college—with its residential exposure to local labor markets, amenities, or *hukou* barriers—reshapes graduates’ perceptions of a location’s benefits and costs.²

Challenging conventional wisdom, we find a negative “general human capital” effect: conditional on enrollment location, a one-percentage-point increase in college attendance probability lowers the post-graduation migration probability by approximately 0.2 percentage points ($p < 0.05$). This “local anchoring” effect suggests that the commonly observed mobility premium for graduates stems almost entirely from the experience of leaving home for college, not from the degree itself. We interpret this finding as a Local Average Treatment Effect (LATE) for the marginal students whose enrollment was induced by the expansion. For this specific group, a degree strengthens local labor market attachment—a rational response within institutional contexts, common in developing economies, where the returns to human capital are powerfully shaped by local social norms and institutional frictions.

This study contributes to the literature on the spatial sorting of skilled workers by providing causal estimates that disentangle the distinct effects of college education and enrollment location on migration (e.g., Venables, 2011; Eeckhout, Pinheiro, and Schmidheiny, 2014; Diamond, 2016; Handbury, 2021; Gaubert and Robert-Nicoud, 2022; Olney and Thompson, 2024).

In doing so, we offer two key insights. First, we identify a causal effect of college location on subsequent residency (Bound et al., 2004; Groen, 2004; Abel and Deitz, 2012; Fitzpatrick and Jones, 2016; Winters, 2020; Huang, Xing, and Cui, 2022; Amornsiripanitch et al., 2023), thereby showing how the geographic distribution of universities can shape youth mobility and spatial inequality (Kelchtermans and Verboven, 2010; Gibbons and Vignoles, 2012; Fu et al., 2022; Fabre, 2023; Ishimaru, 2023). Second, our finding of a net negative effect of college attainment on mobility challenges the conventional view (Wozniak, 2010; Machin, Pelkonen, and Savanes, 2012; Malamud and Wozniak, 2012; Grogger and Hanson, 2015; Weiss, 2015; Kerr et al., 2017; Ding, 2021), suggesting that local investment in higher education may anchor talent rather than induce brain drain in certain institutional contexts.

²The *hukou* system is a household registration system implemented in mainland China. It officially designates an individual as a permanent resident of a specific area. A household registration record contains essential identifying information such as the person’s name, date of birth, and place of permanent residence. See Section 6.3 below for more details. For comprehensive reviews of the *Hukou* system in China, refer to Cheng and Selden (1994), Chan (2015), and Young (2013).

Methodologically, our study joins a growing literature that exploits natural experiments to identify the causal drivers of migration and its consequences (Edwards and Ureta, 2003; Yang, 2008; Malamud and Wozniak, 2012; Aydemir, Kirdar, and Torun, 2021; McKenzie and Yang, 2024; Fan and Li, 2023; Howell, 2023, 2024; Allende, Luksic, and Navarrete, 2024). While prior work has exploited shocks such as remittance flows, military drafts, or school reforms, we leverage a large-scale expansion of higher education as an exogenous source of variation in both educational attainment and location choice, allowing for a clean estimation of their separate causal effects.

The paper proceeds as follows. Section 2 reviews the institutional background. Section 3 describes our data and measures. Section 4 outlines our empirical strategy. Section 5 presents the main empirical findings. Section 6 explores the underlying mechanisms. Section 7 concludes.

2 Institutional Background

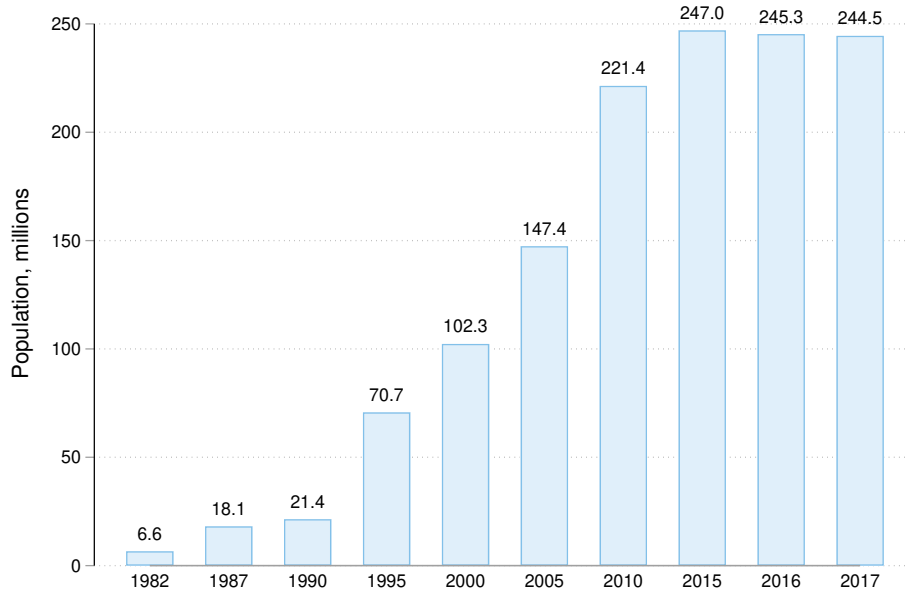
2.1 Migration Patterns and Youth Mobility

China’s economic transition has driven a sharp rise in internal migration. The migrant population grew from 6.6 million in 1982 to more than 200 million by 2010, peaked around 2015, and has remained elevated since (Figure 1). This rise in mobility followed a sequence of major economic reforms. The shift began with the 1979 “household responsibility system,” which increased agricultural productivity and freed rural labor. The rise of Township and Village Enterprises (TVEs) in the mid-1980s further accelerated this shift by increasing urban labor demand.³ China’s 2001 WTO accession and rapid urbanization then intensified migration.

A prominent feature of this modern migration landscape is the high mobility of young adults. According to the 2010 census, approximately 22% of individuals aged 18 to 24 migrated that year. This large-scale movement is a rational response to evolving geographic differences in skill demand, wages, housing costs, amenities, and mobility restrictions (Diamond and Gaubert, 2022).

³TVEs in China are market-oriented public enterprises administered by local governments and primarily located in townships and villages. See Li (1993) for a comprehensive introduction of TVEs in China.

Figure 1—China’s Migration Population, 1982-2017



Notes: This figure plots China’s migrant population from 1982 to 2017. Data are obtained from the 2018 China Migration Population Development Report.

2.2 The College Admissions and Provincial Quota System

China’s higher education system blends centralized governance with local discretion. Most universities are public and heavily state-funded (Fang and Liu, 2018; Liu et al., 2020; Fedasiuk, Loera Martinez, and Puglisi, 2022; Fan and Li, 2024). Provincial and city governments deploy decentralized policies—such as financial aid, institutional subsidies, and infrastructure investment—to expand access and retain talent.

The principal instrument of central control is the admissions quota system, reinstated alongside the National Higher Education Entrance Examination (*Gaokao*) in 1977 (Yang, 2021). Each year, the Ministry of Education, together with provinces and universities, sets institution-specific enrollment quotas (Bollinger, Ding, and Lugauer, 2022; Feng and Xia, 2022). Crucially, quotas are province-specific. Each institution partitions its seats into an “in-province” allotment, reserving the majority of seats for local students, and multiple “out-of-province” allotments earmarked for particular sending provinces. For example, a university in Jiangsu reserves a large share for Jiangsu candidates and smaller, separate quotas for candidates from Sichuan.

Although matching procedures (e.g., sequential vs. parallel) vary across provinces (Chen,

Jiang, and Kesten, 2020), the core structure is invariant in that applicants primarily compete with peers from their home province for quota-limited seats. This institutional structure forms the foundational premise for our identification strategy.

2.3 The 1999 Higher Education Expansion

Following a period of modest recovery after the Cultural Revolution, with annual college enrollment growing from 0.4 million to 1 million between 1978 and 1998, China's higher education system was radically transformed in 1999. In response to slowing economic growth after the 1997 Asian Financial Crisis and rising youth unemployment, the State Council initiated an abrupt, top-down expansion of college enrollment, tripling the target gross college admission rate from 1.5 to 4.5 percent. The implementation was sudden: in June 1999, just two weeks before the national college entrance exam, the MOE announced a 38% increase in the national seat quota, a sharp deviation from the existing five-year plan.⁴ In subsequent years, the MOE continued to implement substantial annual enrollment increases, while maintaining the benchmark enrollment quota originally established by the State Council.⁵

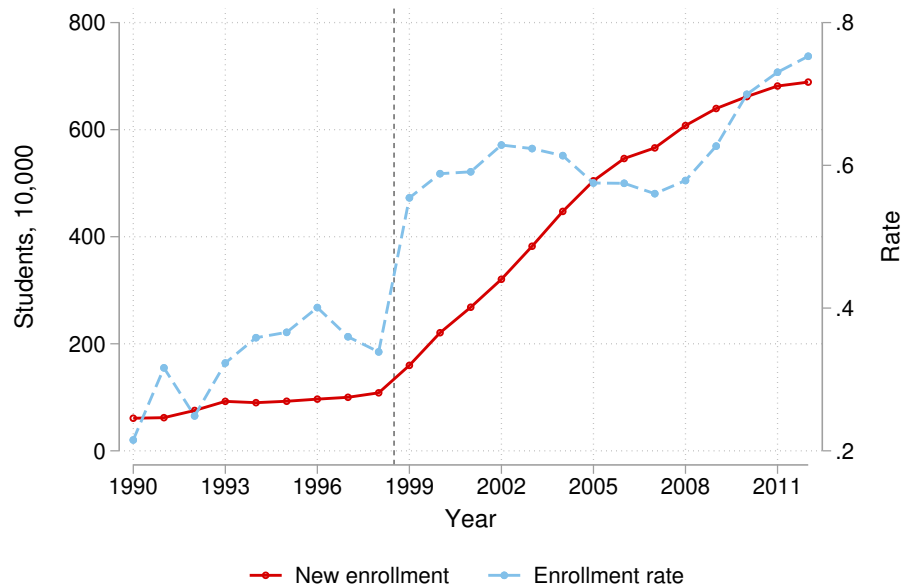
Figure 2 illustrates the policy's dramatic impact. After years of stability at roughly one million students, annual college enrollment jumped by over 40% in 1999, creating a sharp discontinuity (Li, Whalley, and Xing, 2014). This steep upward trend continued, with new enrollments reaching nearly 7 million by 2012. The enrollment rate shows a similar trend, surging from approximately 30% before the expansion to nearly 80% by 2012.

Crucially, this national trend masks significant heterogeneity in the expansion's intensity across provinces. Figure 3 illustrates this by plotting enrollment trends for selected provinces, showing that while all experienced growth after 1999, the magnitude and trajectory of the increase varied substantially. The geographic distribution of this variation is shown in Figure 4, which maps the change in the college enrollment rate between 1998 and 2010. The map reveals a distinct spatial pattern, highlighting the substantial spatiotemporal variation that is the key feature of the natural experiment our study exploits.

⁴Throughout the remainder of this paper, we use the terms "higher education expansion," "college seat expansion," and "college expansion" interchangeably to refer to the growth of college seats and enrollment.

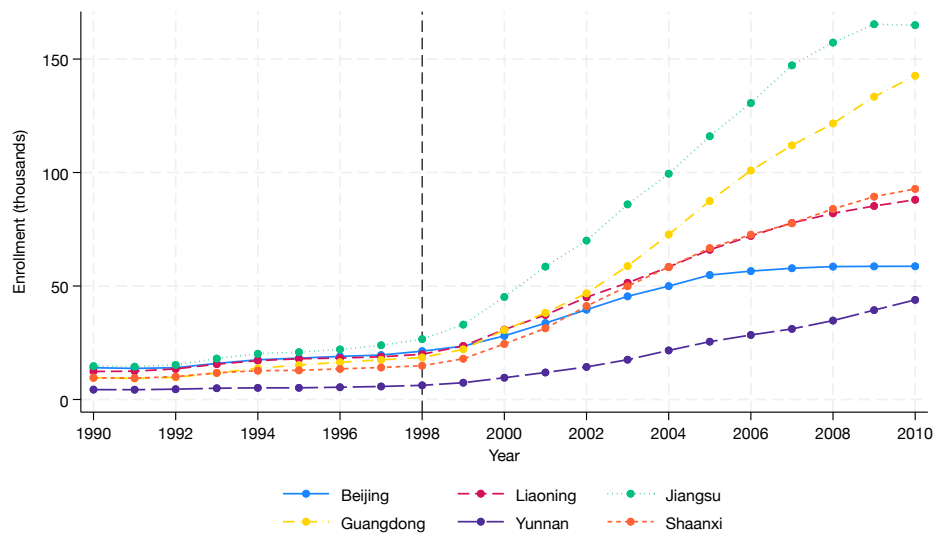
⁵For more details on the policy change background, please refer to the discussions in Feng and Xia (2022).

Figure 2—The Expansion of Higher Education in China



Notes: This figure presents the trends in national college enrollment and the corresponding enrollment rate, calculated as the ratio of college enrollments to high school graduates. Data for this analysis are from the *Educational Statistics Yearbooks of China* and the National Bureau of Statistics.

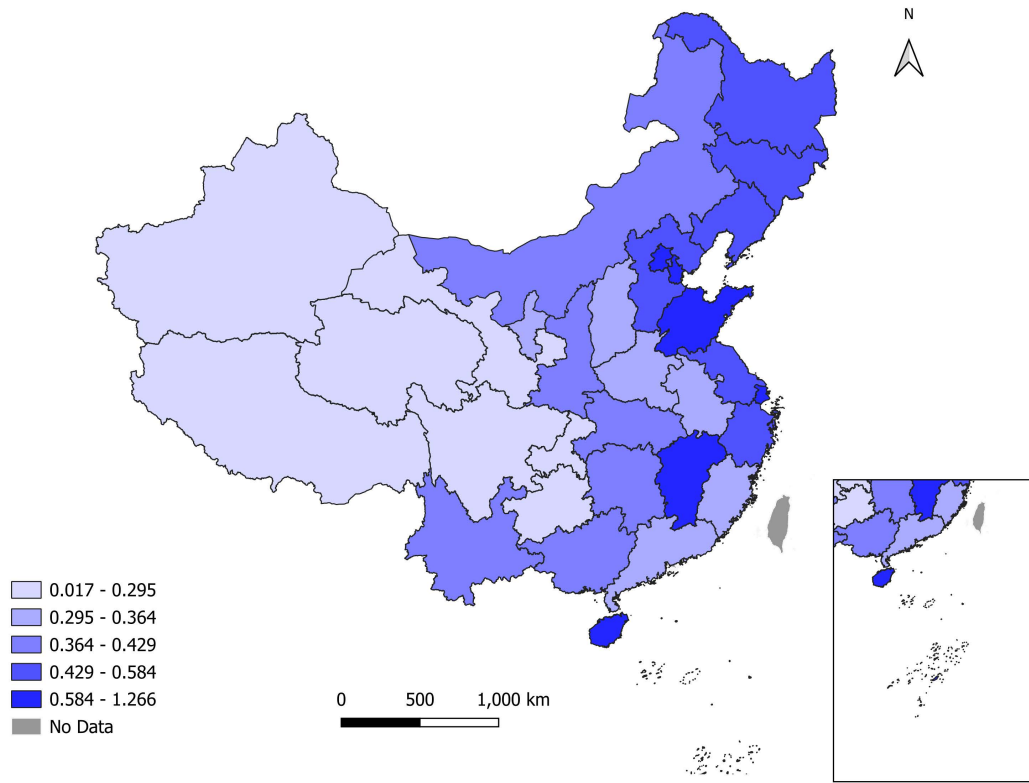
Figure 3—The Expansion of Higher Education in China across Provinces



Notes: This figure presents the trends in college enrollment for selected provinces. Data for this analysis are from the *Educational Statistics Yearbooks of China* and the National Bureau of Statistics.

This cross-province variation arose because the allocation of new quotas was not based on a forward-looking economic formula but was instead driven by administrative discretion and

Figure 4—Enrollment Rate Change by Province



Notes: This map shows the change in enrollment rate, measured by college enrollment divided by high school graduates, for each province between 1998 and 2010. Data for this analysis are from the *Educational Statistics Yearbooks of China* and the National Bureau of Statistics.

historical path-dependency. To achieve rapid scale, the MOE allocated the majority of new seats to large, existing universities, which were already geographically concentrated due to the legacy of the “1952 System” (Glaeser and Lu, 2018). This process was further shaped by a political economy dimension: local governments bargained for an increased share of admission slots for their homegrown students in exchange for providing resources, such as land and funding.

Importantly, this allocation process, implemented through the province-specific quota system, created two distinct, plausibly exogenous shocks for students in any given province: a shock to the supply of “in-province” seats at local universities, and a separate shock to the supply of “out-of-province” seats at universities in other regions. While not purely random at the regional level, these shocks—originating from a macroeconomic shock and subsequent policy response—are unlikely to be correlated with the unobserved, individual-level determinants of our cohorts’ migration

decisions, especially after adjusting for differences in individual and regional characteristics.

3 Data and Measures

3.1 Population Census

Our empirical analysis uses microdata from the 2010 Chinese Population Census, which provides detailed individual-level information, including birthplace, residence five years prior, and current residence, alongside standard demographic characteristics. This allows us to construct our primary outcome: a binary indicator for cross-province migration, which equals one if an individual has resided in a province different from their birth province for at least six months. Our analytical sample is restricted to individuals with at least a high school education, the population directly affected by the college expansion policy. To isolate physical migration from administrative changes, we further exclude individuals whose *hukou* registration changed without a corresponding move across provincial borders. Appendix Section A.1 provides complete details on the sample construction.

3.2 College Enrollment Location

A central challenge for our analysis is identifying college enrollment locations, as this information is not explicitly recorded in the population census. We overcome this by developing an inference method that leverages the 2010 census's five-year retrospective residential history. The 2010 census, conducted in November, records each individual's place of residence in both October 2010 and October 2005. Specifically, we use an individual's reported residence in October 2005 as a proxy for their college location.

This method is valid for the specific cohorts who were likely to be enrolled at that time. Assuming a standard college entry age of 18 (Wang, 2010), our target sample consists of individuals who matriculated between September 2003 and September 2005.⁶ Earlier cohorts are excluded as

⁶To estimate an individual's year of college matriculation, we apply the standard school entry age rule in China. Due to the September 1st birth date cutoff for starting school, individuals born on or after this date are typically one year older than their classmates born in the same calendar year. Accordingly, we estimate the matriculation year as (Birth Year + 19) for those born on or after September 1st, and as (Birth Year + 18) for those born before this date.

they may have already graduated and relocated by the October 2005 snapshot, while later cohorts are also excluded as they had not yet started college.⁷

By comparing an individual’s inferred college province (residence in 2005) with their birth province, we construct our key variable for in-province versus out-of-province enrollment. This procedure yields a sample of young professionals in the early stages of their careers as of 2010, a key demographic for studying labor market sorting and the spatial allocation of human capital (Topel and Ward, 1992; Fan and Zou, 2021).

3.3 Location-Specific Characteristics

To account for location-specific heterogeneity, our analysis incorporates detailed sub-provincial economic characteristics. A key prerequisite for this approach is linking individuals to their birth cities, but the census does not directly report this information. To overcome this, we develop a procedure to infer birth city using information on birth province and *hukou* registration (detailed in Appendix Section A.2).

Using these inferred birth cities, we merge individual records with a rich set of city-level characteristics, drawn from the *China City Statistic Yearbooks*, such as population, GDP, and employment. To capture broader regional trends, we also incorporate provincial-level data. Our final analytical sample consists of 63,498 individuals, representing 18.9% of the census population aged 23 to 26.

A potential concern is that our inference procedure, which excludes individuals whose birth city cannot be determined, may introduce selection bias. To assess this, we conduct a balance test by regressing key observable characteristics on an indicator for being dropped from the sample. The results, presented in Table A.1, show no systematic differences between our analytical sample and the excluded individuals. The coefficients on key individual and regional characteristics are statistically insignificant and economically small, providing confidence that our sample construction methodology does not introduce significant selection bias.⁸

⁷For instance, students who began in 2002 might have completed a three-year vocational degree (Wu and Ye, 2018) and moved before October 2005, while the 2001 cohort would have graduated in June 2005, potentially relocating before the census snapshot.

⁸While the age difference is marginally statistically significant (0.007 years), its magnitude (less than 3 days) is economically negligible.

3.4 Summary Statistics

Table 1 presents summary statistics for our analytical sample, with Panel A and Panel B focusing on individual-level characteristics and regional factors, respectively. It reports mean values and standard deviations for three groups: individuals without a college degree ($College = 0$), those with a college degree ($College = 1$), and the subset of graduates who attended college outside their birth province ($CollegeOut = 1$). The raw data show that a slightly higher share of non-college-educated individuals are migrants (12.2%) compared to their college-educated peers (8.9%). However, a striking 91% of those who attended college out-of-province are migrants. This strong raw correlation provides preliminary evidence that enrollment location, rather than education itself, may be a key predictor of migration.

The groups also differ along several observable characteristics. Compared to the college-educated group, which has a balanced gender ratio, non-college-educated individuals and those who studied out of province are more likely to be male. While ethnicity and age are similar across groups, individuals without a college education tend to have larger households. Both non-college-educated individuals and those who attended college out-of-province tend to originate from cities and provinces that are, on average, less economically developed and farther from the coast. We control for these covariates valued at the time of the individual's college matriculation year in subsequent analyses to account for these observed differences in pre-determined characteristics.

4 Empirical Strategy

This section outlines our empirical strategy. We first present a baseline model linking migration to college choices and discuss the inherent identification challenges. We then detail our IV approach designed to estimate the causal effects of interest.

Table 1—Summary Statistics

	(1) <i>College=0</i>	(2) <i>College=1</i>	(3) <i>CollegeOut=1</i>
<i>Panel A. Individual Characteristics</i>			
Migrant	0.122 (0.327)	0.089 (0.284)	0.910 (0.286)
<i>College</i>	0.000 (0.000)	1.000 (0.000)	1.000 (0.000)
<i>CollegeOut</i>	0.000 (0.000)	0.091 (0.288)	1.000 (0.000)
Male	0.541 (0.498)	0.484 (0.500)	0.562 (0.496)
Han ethnicity	0.964 (0.185)	0.963 (0.189)	0.954 (0.210)
Age	24.081 (0.915)	24.135 (0.910)	24.142 (0.919)
Household size	4.221 (1.956)	3.699 (1.805)	3.964 (1.600)
<i>Panel B. City and Province Characteristics</i>			
City population (million)	6.203 (4.791)	6.339 (4.565)	6.120 (5.149)
City GDP per capita (10,000 RMB)	1.469 (1.172)	1.884 (1.441)	1.195 (0.881)
City tertiary employment share (%)	36.695 (7.503)	38.179 (8.326)	36.442 (6.535)
City distance to coast (100 km)	3.964 (3.434)	3.435 (3.392)	4.425 (3.338)
Province urban employment (million)	0.176 (0.067)	0.193 (0.083)	0.165 (0.054)
Province GDP per capita (10,000 RMB)	1.405 (0.765)	1.603 (0.962)	1.197 (0.585)
Provincial capital city	0.127 (0.333)	0.174 (0.379)	0.106 (0.308)
Municipality	0.060 (0.237)	0.101 (0.302)	0.043 (0.203)
Province GDP growth rate	0.172 (0.038)	0.171 (0.038)	0.169 (0.039)
Province employment growth rate	0.055 (0.048)	0.053 (0.051)	0.049 (0.041)
Observations	32,118	31,380	2,858

Note: This table presents the summary statistics for the main sample. Mean values are reported, with standard deviations in parentheses.

4.1 Baseline Specification and Identification Challenges

We begin by specifying a linear probability model that relates post-graduation migration to college enrollment decisions:

$$M_{ijt} = \beta_0 + \beta_1 \text{College}_i + \beta_2 \text{CollegeOut}_i + \beta_3 \mathbf{X}_{ijt} + \phi_j + \phi_t + \varepsilon_{ijt}, \quad (1)$$

where the dependent variable, M_{ijt} , is a binary indicator equal to one if individual i from birth cohort t and birth province j resides outside their birth province. The key explanatory variables are College_i , which equals one if individual i attended college (regardless of their college enrollment location), and CollegeOut_i , which equals one if individual i attended an out-of-province college. This specification allows us to disentangle the effect of acquiring a college education (β_1) from the effect of the out-of-province location experience (β_2).⁹

The vector $\mathbf{X}_{ijt}$ controls for a rich set of pre-determined controls, including personal characteristics (gender, ethnicity, age and its square, and household size), birth-city attributes measured at age 18 for individual i (population, GDP per capita, employment composition, proximity to the coast), and time-varying birth-province characteristics (GDP per capita, urban employment, and their growth rates).¹⁰ A full list of covariates is available in Table 1. We also include birth-province fixed effects (ϕ_j) and birth-cohort fixed effects (ϕ_t) to account for unobserved time-invariant regional differences and national cohort-level trends. Standard errors are clustered at the birth-province level to account for arbitrary within-province correlation in the error term, ε_{ijt} .

OLS estimates of equation (1) are valuable for capturing associations but face considerable challenges in establishing causality. The primary concern is that college attendance and location choices are likely correlated with unobserved factors—such as innate ability, ambition, or the

⁹The coefficient β_1 measures the effect of college education *per se* (general human capital), representing the difference in migration propensity between an in-province college graduate and a non-college-educated individual: $M(\text{College} = 1, \text{CollegeOut} = 0) - M(\text{College} = 0, \text{CollegeOut} = 0) = (\beta_0 + \beta_1) - \beta_0 = \beta_1$. The coefficient β_2 isolates the effect of the enrollment location experience, representing the additional effect of attending college out-of-province relative to in-province: $M(\text{College} = 1, \text{CollegeOut} = 1) - M(\text{College} = 1, \text{CollegeOut} = 0) = (\beta_0 + \beta_1 + \beta_2) - (\beta_0 + \beta_1) = \beta_2$.

¹⁰Two controls warrant clarification. First, while birth-cohort fixed effects account for common shocks affecting an age group, we also control for individual age and its squared term to capture any remaining non-linear age effects on migration decisions within each cohort. Second, our controls include an indicator for birth in a provincial capital or a direct-administered municipality (*Zhixiashi*), a city with provincial-level status (e.g., Beijing, Shanghai, Tianjin, and Chongqing during our study period).

strength of hometown ties—that also directly influence migration decisions. Furthermore, reverse causality is a concern, as migration prospects themselves may influence educational investments (Navarro and Zhou, 2024). While our extensive set of controls and fixed effects account for many observed and unobserved confounders, they cannot fully address these unobserved sources of endogeneity. Therefore, we employ an IV strategy to isolate a more credible causal effect.

4.2 Instrumental Variable (IV) Construction

To identify the causal effects, we exploit the plausibly exogenous variation in college access generated by China’s massive college expansion in the late 1990s. This policy was implemented through a province-specific quota system, creating geographic and temporal variation in the supply of college seats that provides a powerful natural experiment (e.g., Feng and Xia, 2022).

We construct two instruments corresponding to our two endogenous variables. These instruments measure the province-specific intensity of the in-province and out-of-province college expansion. First, using the 2000 population census, we calculate the initial share of college students from each origin province j that were enrolled in each destination province k in the base year $t_0 = 2000$ (s_{jk,t_0}).¹¹ The calculation follows the equation:

$$s_{jk,t_0} = \frac{N_{jk,t_0}}{N_{k,t_0}} \quad (2)$$

where t_0 is the initial period—Year 2000, which follows the significant expansion of higher education in 1999. N_{k,t_0} denotes the total number of college enrollments in destination province k during the initial period, while N_{jk,t_0} represents the total number of college enrollments from origin province j to destination province k during the same initial period. This equation allows us to derive a comprehensive set of origin-destination pairs containing the shares s_{jk,t_0} .

We then interact these historical shares with the annual college enrollment in each destination province (E_{kt}) and normalize by the number of high school graduates in the origin province (H_{jt}) in year t , following the equations below:

$$IR_{jt} = \frac{E_{jt} \times s_{jj,t_0}}{H_{jt}}, \quad (3)$$

¹¹This calculation uses the 2000 census, as it is the earliest survey with detailed birthplace information.

$$OR_{jt} = \frac{\sum_{k \neq j} E_{kt} \times s_{jk,t_0}}{H_{jt}}, \quad (4)$$

This yields our two instruments: the predicted in-province enrollment rate (IR_{jt}) and out-of-province enrollment rate (OR_{jt}) for each province j in year t , which capture the supply-side shocks to local and distant educational opportunities for each province-cohort.

Figure 5 visualizes these constructed shocks for 2005, revealing substantial and distinct geographic variation across China. The expansion of in-province enrollment (IR_{jt}) was concentrated in Beijing, Tibet, and the northeastern and southeastern coastal provinces (Figure 5a). In contrast, the expansion of out-of-province enrollment (OR_{jt}) was more pronounced for students from peripheral and inland provinces (Figure 5b). This rich spatiotemporal variation in exposure to the two types of expansion shocks provides the crucial variation for identification.

4.3 Identification Strategy, Estimation, and Instrument Validity

Our estimation strategy addresses the binary nature of the endogenous college choice variables ($College_i$ and $CollegeOut_i$) using a two-step procedure proposed by Wooldridge (2010, Chapter 18, Procedure 18.1). In the first step, we estimate separate probit models for college attendance and out-of-province enrollment. Each choice is regressed on our instruments (IR_{jt} and OR_{jt}), measured at age 18 for individual i , along with a full set of exogenous controls and fixed effects:

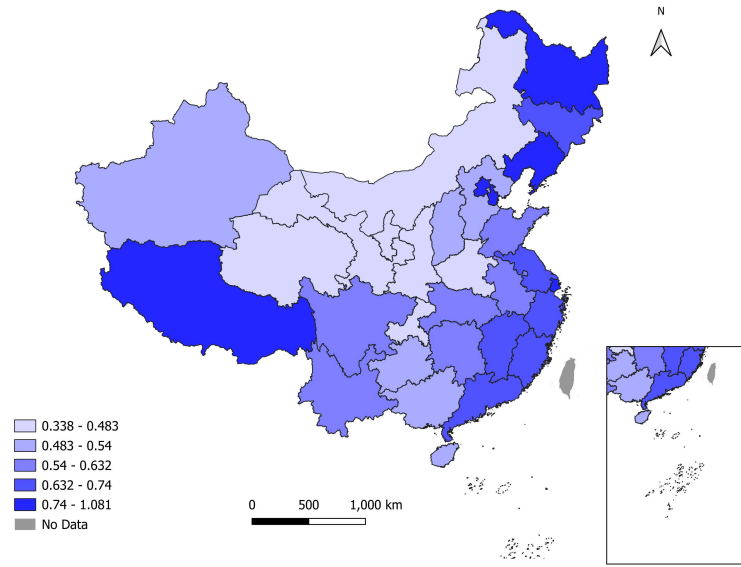
$$College_{ijt} = \Phi(\alpha_0 + \alpha_1 IR_{jt} + \alpha_2 OR_{jt} + \alpha_3 \mathbf{X}_{ijt} + \eta_j + \eta_t + u_{ijt}), \quad (5)$$

$$CollegeOut_{ijt} = \Phi(\theta_0 + \theta_1 IR_{jt} + \theta_2 OR_{jt} + \theta_3 \mathbf{X}_{ijt} + \gamma_j + \gamma_t + v_{ijt}), \quad (6)$$

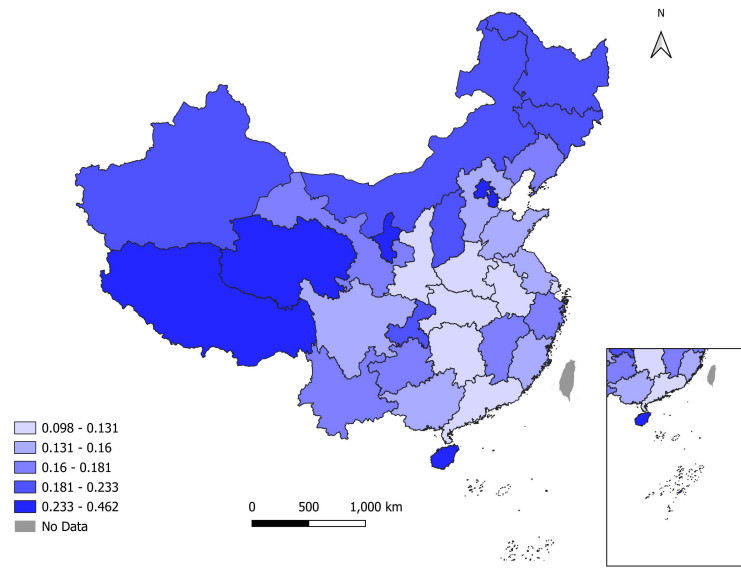
In the second step, the predicted probabilities, $\widehat{College}$ and $\widehat{CollegeOut}$, serve as instruments for the actual choices in a two-stage least squares (2SLS) estimation. This approach has key advantages over a standard 2SLS framework that directly uses the aggregate shocks (IR and OR) as instruments, as it accommodates a nonlinear and heterogeneous response of individual choices to the province-level policy shocks and yields consistent estimates under the robust inference even if the first-stage probit models are misspecified (Wooldridge, 2010, pp. 622–624).

Since our instrumental variables vary at the province-cohort level while our outcome is measured at the individual level, we weight each observation by the number of high school

Figure 5—College Enrollment Rates Predicted by In- and Out-of-Province College Seat Expansions



(a) In-province college enrollment rate, 2005



(b) Out-of-province college enrollment rate, 2005

Notes: This figure shows the predicted in-province (IR_{jt}) and out-of-province (OR_{jt}) enrollment rate for each province in 2005. The rates are constructed using equations (3) and (4) with base-year shares $s_{jk,2000}$ from the 2000 census and annual data from the *Educational Statistics Yearbooks of China* and the National Bureau of Statistics.

graduates in the individual’s province-cohort. This standard procedure for regressions with mixed-level data addresses potential heteroskedasticity by ensuring that each province-cohort’s contribution to the estimate is proportional to its size, effectively down-weighting the influence of smaller groups whose aggregate measures are estimated with higher variance.

The validity of our IV strategy hinges on the exclusion restriction that our instruments, conditional on controls, affect migration only through their influence on college attendance and location. Several strands of evidence support this assumption.

First, the institutional context of the 1999 expansion is favorable. The policy was an abrupt, unanticipated response to a macroeconomic shock, not a reaction to regional growth trends. Moreover, the allocation of new seats was driven by a combination of historical path-dependency and political economy, rather than a province’s future migration potential. Specifically, the distribution relied on the pre-determined capacity of existing universities and was shaped by a bargaining process where provincial governments traded fiscal support for more local admission slots. This process introduced variation tied to local fiscal capacity, not the unobserved migration potential of a province’s youth. Our use of pre-determined 2000 enrollment shares further ensures that later migration trends do not contaminate our instruments.

Second, we provide direct empirical support for the exogeneity of the shocks. An event-study analysis, presented in Figure A.1, shows no evidence of differential pre-trends between high- and low-expansion provinces in key economic outcomes before the policy’s implementation. Furthermore, as shown in Table A.2, we find no systematic relationship between our instruments and other time-varying provincial characteristics that could be correlated with migration decisions. The absence of correlation with these observables strengthens our confidence that the instruments are also orthogonal to unobserved confounders (Oster, 2019). Taken together, this evidence provides strong support for our identification strategy.

5 Baseline Results

5.1 First-Stage Results: The Effect of Expansion on Enrollment Choice

This section presents the first stage of our two-step IV analysis, estimating the impact of in-province and out-of-province college expansion shocks on individuals' educational choices. Table 2 reports the raw estimates and the marginal effects from the probit models specified in equations (5) and (6). Column 1 examines the decision to attend college at all (*College*), while columns 2 and 3 analyze the choice to enroll outside one's home province (*CollegeOut*).

The results in column 1 show that shocks to the supply of college seats predict overall college attendance. Based on the marginal effect estimates, a one-unit increase in the in-province expansion instrument (*IR*)—conceptually equivalent to an entire cohort of a province's high school graduates being offered a local college seat—increases a typical high school graduate's probability of attending college by 12.6 percentage points, an effect that is both statistically and economically significant. The out-of-province expansion shock (*OR*) has a positive but less precisely estimated effect.

Columns 2 and 3 show distinct effects on out-of-province enrollment. Additional in-province seats (*IR*) slightly reduce this probability, but the estimates are statistically insignificant. By contrast, more out-of-province seats (*OR*) significantly increase it: in the full sample (column 2), a one-unit rise in *OR* raises the probability by 2.2 percentage points; among college attendees (column 3), the marginal effect is 42.5 percentage points. These patterns indicate strong, differential first-stage relevance for the two endogenous variables.

5.2 Causal Effects of Education and Enrollment Location on Migration

Table 3 reports 2SLS estimates of equation (1), instrumenting for *College* and *CollegeOut* with $\widehat{College}$ and $\widehat{CollegeOut}$. Panel A summarizes the first-stage (columns 1 and 2) and reduced-form (column 3) results. Instrument strength, assessed by the Kleibergen–Paap rk Wald *F*-statistic (Andrews, Stock, and Sun, 2019; Roodman et al., 2019), is 16.82—exceeding conventional thresholds and thus alleviating weak instrument concerns. Columns 1 and 2 demonstrate that the predicted probabilities strongly predict the corresponding enrollment choices.¹² Column 3 reports direct intention-to-treat effects: the college attendance instrument predicts lower migration, whereas

¹²The opposite signs across instruments reflect the mechanical negative correlation between *College* and *CollegeOut* (e.g., in-province attendance implies *College* = 1 and *CollegeOut* = 0)

Table 2—The Effect of College Expansion on College Attendance Choices, Probit Estimates

	(1) <i>College</i>	(2) <i>CollegeOut</i>	(3) <i>CollegeOut</i>
In-province college expansion (<i>IR</i>)	0.331* (0.199)	-0.274 (0.210)	-0.423 (0.259)
Out-of-province college expansion (<i>OR</i>)	0.128 (0.544)	2.396* (1.383)	2.805* (1.617)
Han ethnicity	-0.080*** (0.025)	-0.013 (0.031)	0.016 (0.033)
Male	-0.138*** (0.016)	0.088*** (0.022)	0.179*** (0.023)
Household size	-0.084*** (0.005)	-0.009 (0.007)	0.031*** (0.008)
Age	0.544 (0.470)	-1.333** (0.674)	-2.042*** (0.737)
Age square	-0.011 (0.010)	0.027* (0.014)	0.041*** (0.015)
City population	-0.012*** (0.004)	-0.002 (0.009)	0.004 (0.009)
City GDP per capita	0.106*** (0.014)	-0.093*** (0.025)	-0.145*** (0.031)
City tertiary employment share	0.004** (0.002)	0.002 (0.002)	0.001 (0.003)
City distance to the coast	-0.022** (0.011)	0.005 (0.021)	0.014 (0.023)
Province urban employment	-0.319 (0.977)	-1.941 (1.504)	-1.615 (1.761)
Province GDP per capita	-0.078 (0.050)	0.252*** (0.080)	0.269*** (0.097)
Provincial capital city	0.121*** (0.036)	-0.154** (0.067)	-0.255*** (0.076)
Municipality	-0.019 (0.251)	0.226 (0.556)	0.199 (0.488)
Province GDP growth rate	0.176 (0.216)	0.341 (0.299)	0.439 (0.337)
Province employment growth rate	-0.007 (0.162)	-0.210 (0.182)	-0.237 (0.187)
Observations	63,498	63,498	31,380
Marginal effect of <i>IR</i>	0.126* (0.076)	-0.025 (0.019)	-0.064 (0.039)
Marginal effect of <i>OR</i>	0.049 (0.207)	0.022* (0.126)	0.425* (0.245)

Note: This table reports results from probit models estimating the effect of college expansion shocks on educational choices, using data from the 2010 census. The dependent variable in column 1 is a dummy variable for college attendance. In columns 2 and 3, the dependent variable is a dummy variable for attending college outside one's birth province; the sample for column 3 is restricted to college-educated individuals. All specifications include birth province and birth-cohort fixed effects and are weighted by the number of high school graduates in the individual's province-cohort. Robust standard errors are clustered at the birth province level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

the out-of-province instrument predicts higher migration.

Table 3—Estimating the College Education and Enrollment Location Effects, 2SLS

Dependent variable:	(1) <i>College</i>	(2) <i>CollegeOut</i>	(3) Migration
<i>Panel A: First-stage & Reduced-form estimates</i>			
$\widehat{College}$	2.114*** (0.215)	-0.131** (0.063)	-0.792*** (0.193)
$\widehat{CollegeOut}$	-1.009* (0.523)	1.133*** (0.187)	3.601*** (0.481)
Observations	63,498	63,498	63,498
First-stage <i>F</i> -statistic	16.82	16.82	-
Dependent variable: Migration	IV	OLS	
<i>Panel B: Second-stage estimates</i>			
<i>College</i>	-0.189** (0.0735)	-0.102*** (0.0178)	
<i>CollegeOut</i>	3.011*** (0.485)	0.859*** (0.0143)	
Observations	63,498	63,498	

Notes: This table presents the main estimation results of equation (1). Columns 1 and 2 in panel A show the first-stage results of 2SLS estimation, and column 3 reports the reduced-form estimates. Panel B presents the second-stage IV estimates (column 1) and a comparative OLS estimate (column 2). The dependent variable in panel B and for the reduced-form estimates (column 3) is a binary indicator for postcollege migration. All regressions control for birth-province fixed effects, birth-cohort fixed effects, and individual and location characteristics. Regressions are weighted by the number of high school graduates. Robust standard errors, clustered at the birth province level, are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Column 1 of panel B reports our main causal estimates. Consistent with the reduced-form estimates, the results reveal two distinct and contrasting effects. First, we find a negative “general human capital” effect: conditional on enrollment location, a one-percentage-point increase in the probability of attending college reduces the likelihood of post-graduation migration by about 0.2 percentage points. Second, we find a strong “location experience” effect: a one-percentage-point increase in the probability of enrolling in an out-of-province institution raises postcollege

migration by roughly 3.0 percentage points, net of the education effect.¹³

The Local Average Treatment Effects (LATE) framework, which captures the impact for “compliers” students marginal to college admission expansion, offers a coherent explanation for these results. For graduates who attended college in their home province as a result of college admission expansion, the “local anchoring” effect may arise because their degree—likely from a non-elite institution—carries more weight in the local labor market than nationally, making a stable job at home a rational choice. In contrast, the large location effect of out-of-province enrollment is highly plausible for graduates who attended college out-of-province as a result of college admission expansion, for whom the initial move for education creates powerful path dependence, exposing them to new labor markets and encouraging settlement there.

Finally, column 2 of panel B presents OLS estimates for comparison. While qualitatively similar, the OLS coefficients understate both the negative education effect and the positive location effect, likely due to attenuation bias from measurement error or unobserved heterogeneity. This underscores the importance of our IV strategy.

5.3 Robustness Checks

Our main findings are robust across several sensitivity tests (Table 4). First, addressing model specification, the results are quantitatively similar when estimated without weights (column 1). Second, to account for potentially confounding pre-existing trends, we augment the baseline model with interactions between city-specific linear time trends and local characteristics measured in 1998 (air quality, GDP growth, average years of schooling). Our main coefficients remain qualitatively unchanged (column 2), suggesting the results are not driven by differential pre-expansion trends in local economic characteristics and amenities across cities.

Next, we test the sensitivity of our findings to sample composition. The results are robust to excluding individuals born in first-tier cities (column 3) and provincial capitals (column 4), mitigating concerns that our estimates are driven by sorting into a few dominant urban centers (Xing and Zhang, 2017). To address potential bias from our sample construction due to birth

¹³This large location effect seemingly differs from findings in other contexts, such as the more modest estimates for the U.S. in Winters (2020). However, cross-country comparisons should be made with caution due to institutional, methodological, and population differences.

Table 4—Robustness Checks

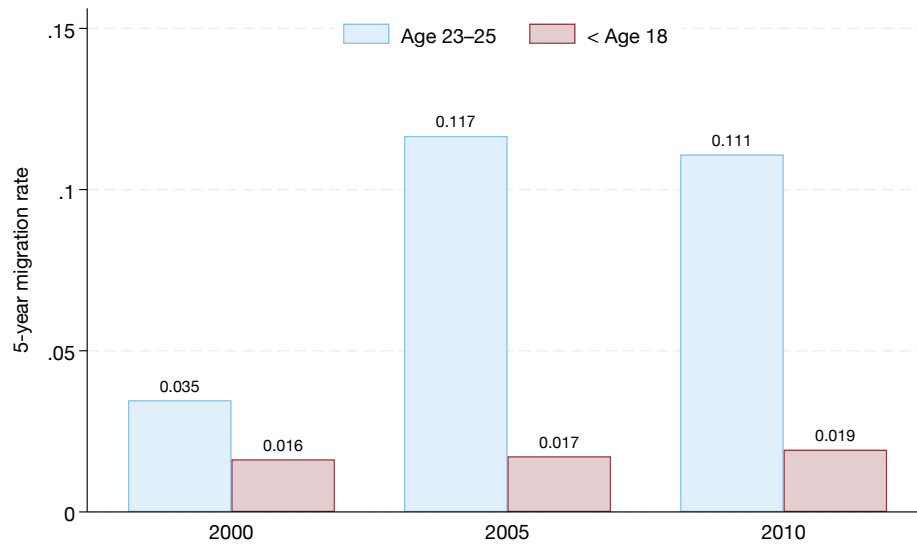
	(1) No Weight	(2) City-specific trend	(3) Drop Tier-1 cities	(4) Drop prov. capital	(5) IPW
College	-0.163** (0.0726)	-0.226*** (0.0633)	-0.200** (0.0881)	-0.253*** (0.0706)	-0.207*** (0.0601)
CollegeOut	3.170*** (0.507)	2.101*** (0.423)	3.122*** (0.598)	2.212*** (0.402)	2.540*** (0.470)
Observations	63,498	63,162	60,058	53,982	63,498
F-statistic	17.09	25.20	12.01	28.42	20.99

Note: This table presents the results of robustness checks using the baseline 2SLS specification. Column 1 is un-weighted. Column 2 adds interaction terms of city-specific trends and pre-trend controls measured in 1998 (average years of schooling, GDP growth rate, and air quality index). Column 3 excludes individuals born in first-tier cities (Beijing, Shanghai, Guangzhou, and Shenzhen). Column 4 excludes individuals born in provincial capitals. Column 5 applies inverse probability weighting (IPW) to account for sample selection. All specifications include the same controls and fixed effects as the baseline model. Robust standard errors, clustered at the birth province level, are in parentheses. $p_{i0.10}$, $p_{i0.05}$, $p_{i0.01}$.

city inference, we implement an inverse probability weighting (IPW) estimation; the coefficients remain consistent with our baseline findings (column 5). Finally, a leave-one-out analysis, which re-estimates the model 28 times while excluding one province at a time, shows that our coefficients are highly stable and not driven by any single province (Figure A.2).

A potential threat to our identification is that individuals who migrated with their families before college age could be misclassified due to a lack of data, creating a spurious correlation between out-of-province enrollment and post-graduation migration. However, descriptive data suggest this channel is unlikely to drive our results. As shown in Figure 6, the five-year cross-provincial migration rate for minors (under 18) was consistently low (around 2-3%) during our period, indicating that pre-college family migration was rare. In contrast, migration rates surge for young adults (ages 23-25) after the typical college graduation age. This confirms that the transition from education to the workforce, not childhood moves, is the primary trigger for long-distance migration in our sample. Therefore, the number of individuals potentially misclassified due to pre-college migration is too small to significantly bias our estimates.

Figure 6—China’s Young Adult and Minor Migration Rate



Notes: This figure plots the 5-year migration rate for young adults (ages 23-25) and minors (under age 18) in censuses of 2000, 2005, and 2010.

6 Explaining the Migration Effects

Our finding that college education per se has a negative impact on migration contrasts with prior research linking higher education to increased mobility ([Malamud and Wozniak, 2012](#); [Weiss, 2015](#)). This result does not imply that college graduates are less mobile overall. Rather, it clarifies that the commonly observed mobility premium for graduates stems almost entirely from the experience of leaving home for college, not from the degree itself. Once this location effect is accounted for, acquiring a college degree in the Chinese context appears to act as a local anchor.

Individuals choose residence by comparing the expected net present value of benefits ([Bilal and Rossi-Hansberg, 2021](#)), which can be shaped by both general human capital acquired through college and location-specific experience gained from living and studying in a place. To understand why a degree—as a signal of general human capital—might anchor an individual locally, we therefore explore several channels through which these factors, operating within China’s institutional and social context, alter the perceived costs and benefits of migrating.

6.1 Returns to Migration (Skill Premium)

Wage differentials are a primary driver of migration (Sjaastad, 1962; Wozniak, 2010), and the potential gains from moving may vary with an individual’s skill level. A college degree raises expected earnings in the home labor market, which reduces the marginal wage incentive to relocate: the relative improvement from an external skilled job shrinks once one’s baseline is higher, particularly after accounting for lost networks, higher living costs, and *hukou* uncertainty. By contrast, for non-college workers, the wage gain from moving from a local low-paid job to a higher-paid job from migration can be large.

To test this income channel, we extend our baseline model in equation (1) by interacting our endogenous treatment variables ($College_{ijt}$ and $CollegeOut_{ijt}$) with a measure of the potential gain in the skilled-wage premium from migration for individuals born in province j (Δw_j). A positive Δw_j indicates that skilled wage premiums are higher in potential destination provinces than in an individual’s home province j (see Appendix Section A.4 for details). To address endogeneity, we instrument for these interaction terms using the interaction terms of Δw_j with predicted probabilities of college attendance and enrollment location.

The estimates in column 1 of Table 5 show that the effect of a college degree on migration turns strongly positive when potential skilled-premium gains are large ($College \times \Delta w$ is positive and significant).¹⁴ This helps reconcile our negative average treatment effect: absent sizeable wage gains from moving, college graduates have little economic incentive to migrate. By contrast, the interaction term $CollegeOut \times \Delta w$ is negative and insignificant, suggesting that the increase in skilled wage premiums from migration has a more pronounced positive effect on migration for those who attend college within their home province, rather than those who attend college outside of it. Non-wage factors, such as personal preferences, amenities, and cultural fit, may matter more for those who have already self-selected into an out-of-province path for college attendance.

¹⁴Although the first-stage F -statistic is below the conventional threshold, the small p -value from the Anderson-Rubin weak IV test alleviates concerns about weak instruments.

Table 5—Heterogeneous Effects of College Attendance and Enrollment Location on Migration

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Skilled wage	Housing price	<i>Hukou</i>	Birth city GDP	Birth city region	Birth city adm. level	Gender
Panel A: Main Effects							
<i>College</i>	-0.134** (0.056)	-0.126* (0.0670)	-0.184*** (0.0680)	-0.278*** (0.103)	-0.299*** (0.0748)	-0.268*** (0.0667)	-0.138** (0.0646)
<i>CollegeOut</i>	2.243** (0.875)	0.454 (0.542)	3.031*** (0.476)	3.650*** (0.952)	3.012*** (0.462)	2.683*** (0.487)	2.997*** (0.483)
Panel B: Interaction Effects							
<i>College</i> × Interaction var.	0.240*** (0.066)	-0.0929*** (0.0200)	-0.0103 (0.0444)	0.238*** (0.0729)	0.286*** (0.0674)	0.265*** (0.0954)	-0.0896*** (0.0278)
<i>CollegeOut</i> × Interaction var.	-1.405 (1.424)	1.105*** (0.358)	0.218*** (0.0841)	-1.349* (0.818)	-1.249*** (0.376)	-0.390 (0.382)	-0.0244 (0.225)
Panel C: Specification Details							
Interaction variable	Δw	Δhp	$\Delta hukou$	$\mathbb{I}[\text{High GDP}]$	$\mathbb{I}[\text{East}]$	$\mathbb{I}[\text{Prov. capital}]$	$\mathbb{I}[\text{Male}]$
Observations	63,498	63,498	63,354	63,498	63,498	63,498	63,498
First-stage F -statistic	5.458	8.280	8.452	3.514	10.18	3.446	8.979
Anderson-Rubin Weak IV	10.52	13.07	9.87	11.81	12.55	11.36	14.11
Anderson-Rubin p -value	0.033	0.011	0.043	0.0188	0.0137	0.0228	0.007

Notes: This table presents results from the heterogeneity analysis. The dependent variable is a binary indicator for out-of-province migration (M_{ijt}). Each column interacts the main treatment variables (*College* and *CollegeOut*) with a different characteristic: The term Δw is the potential migration wage premium for skilled workers in 2005. The term Δhp is the potential migration housing price gap at age 18. The term $\Delta hukou$ measures the difference in *hukou* reform stringency between potential destinations and the origin province at age 18. $\mathbb{I}[\text{High GDP}]$ is a dummy variable equal to one if the individual's birth city province (at age 18) was above the national median. $\mathbb{I}[\text{East}]$ is a dummy variable equal to one if the individual's birth province is in East China (Shandong, Jiangsu, Anhui, Zhejiang, Jiangxi, Fujian, or Shanghai). $\mathbb{I}[\text{Provincial capital}]$ is a dummy variable equal to one if the individual was born in a provincial capital. The term $\mathbb{I}[\text{Male}]$ is a dummy variable equal to one for male individuals. All specifications include the full set of fixed effects (birth-province, birth-cohort) and control variables from the baseline model. Regressions are weighted by the number of high school graduates in the province-cohort. Robust standard errors, clustered at the birth-province level, are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6.2 Sensitivity to Housing Costs

High-productivity locations typically have high housing costs, but migrants' sensitivity to these costs often varies by skill level and the intended duration of their stay. Non-college-educated individuals, frequently engaging in temporary migration due to institutional barriers like the *hukou* system, may be less sensitive to high destination housing costs because they tend to minimize local housing expenditures by using temporary housing accommodation and remitting income home for consumption or investment (Brown and Jimenez-Soto, 2015; Huang and Yi, 2015). This pattern is especially common when children remain in the origin province with relatives (Tao et al., 2015; Ai and Hu, 2016; Gao et al., 2023) and mirrors the “expenditures in the country of origin” channel in the international migration literature, whereby remittances and consumption back home allow migrants to concentrate in high-wage, high-cost cities without being fully burdened by the local cost of living (Albert and Monras, 2022).

We hypothesize that college-educated individuals making permanent migration decisions will be more sensitive to housing prices than non-college-educated individuals, who may be temporary migrants. We test this by interacting our treatment variables with the *ex-ante* housing price gap between potential destinations and an individual's origin province (Δhp_j). A positive Δhp_j indicates that the average housing price in the birth province j is lower than in other potential migration destinations. We construct Δhp_j , as detailed in Section A.5 of the appendix, using housing price data from the year prior to college enrollment. This temporal lag helps ensure that our estimates capture *ex-ante* housing cost expectations and mitigates concerns of reverse causality (Garriga et al., 2021).

The results in column 2 support this hypothesis. College-educated individuals are significantly more deterred by higher housing costs in destination provinces ($College \times \Delta hp_j$ is negative). In contrast, the positive coefficient on $CollegeOut \times \Delta hp_j$ suggests that the out-of-province experience makes graduates more willing to bear high housing costs, likely because they have better information about the associated local benefits, such as wages, career networks, and agglomeration spillovers (Moretti, 2004; De la Roca and Puga, 2017; Charles, Hurst, and Notowidigdo, 2018).

6.3 The Role of *Hukou* Frictions

The *hukou* system, by tying public benefits like healthcare and education to one's place of registration, constitutes a major institutional friction that differentially affects migrants by skill (Tombe and Zhu, 2019; Fang and Huang, 2022; Sieg, Yoon, and Zhang, 2023). A college degree significantly increases a graduate's ability to secure a local *hukou* and access its associated non-wage benefits (e.g., subsidized housing, healthcare, education, talent incentives), effectively turning the degree into a localized asset. This advantage is often strongest in one's home province, where administrative channels and local policies typically favor returning graduates.

We derive two testable implications. First, because students who study away from home often have their *hukou* temporarily transferred to the college enrollment location—providing them with a foothold in a new labor market—we expect the migration effect of out-of-province enrollment to be larger for destinations with more lenient *hukou* policies. Second, because *hukou* rules are location-specific and best understood through direct exposure to a place, the effect of acquiring a degree alone (the general human-capital channel) should not vary systematically with cross-location differences in *hukou* stringency.

To test these hypotheses, we interact our endogenous treatment variables (*College* and *CollegeOut*) with a measure of the difference in *hukou* reform stringency between potential migration destinations and an individual's origin province at age 18 ($\Delta hukou_{jt}$). This measure is based on the *hukou* reform index from Fan (2019), which scores the ease of obtaining a local *hukou* on a scale from 0 to 6 based on news articles, official documents, and government regulations, with higher scores denoting more lenient policies. The construction of $\Delta hukou_{jt}$ is detailed in Section A.6 of the appendix. As the resulting interaction terms are endogenous, we instrument for them using the interactions of the *hukou* reform index with the predicted probabilities of college attendance and out-of-province enrollment from our first-stage models.

The results in column 3 show that while a college degree *per se* does not significantly alter an individual's sensitivity to *hukou* restrictions, the experience of enrolling out-of-province does. The positive and significant coefficient on $CollegeOut \times \Delta hukou$ indicates that those who study away from home are more likely to migrate to destinations with more lenient *hukou* policies. This suggests that their location-specific experience gained during college equips these individuals to

better navigate and benefit from such institutional gateways.

6.4 Heterogeneity by Birth City Characteristics

Columns 4 through 6 explore how the effects vary with the economic characteristics of an individual's birth city. The results consistently show that the "local anchoring" effect of a college degree is significantly weaker for individuals originating from more economically advantaged areas. Specifically, the negative main effect of a college degree is largely offset by positive and significant interaction terms for those born in provinces with above-median GDP (column 4), in the East region (column 5), or in a provincial capital (column 6). This suggests that in places with strong baseline economic opportunities, a college degree provides less of a relative boost to local prospects, thus diminishing its power to anchor graduates.

Simultaneously, the effect of out-of-province enrollment on promoting migration is significantly weaker for individuals from high-GDP provinces (column 4) and the East region (column 5), as indicated by the negative interaction terms. This is likely because the marginal benefit of migrating to another prosperous region is smaller for those already originating from such regions. Interestingly, the effect of out-of-province enrollment does not significantly differ based on whether one was born in a provincial capital (Column 6).

The weaker anchoring effect for provincial capitals appears inconsistent with the hypothesis of a stronger anchor where the *hukou* is most valuable. This inconsistency likely arises because the binary indicator $\mathbb{I}(\text{Prov. capital})$ captures a bundle of characteristics beyond just *hukou* value. Provincial capitals offer superior baseline labor market opportunities even without a degree, potentially reducing the relative gain from obtaining one locally. Furthermore, individuals born in these major hubs may have greater exposure to national opportunities and possess unobserved characteristics (e.g., family background favoring mobility) that attenuate local anchoring effects. The overall economic dynamism and connectivity of the capital city seem to outweigh the *hukou*'s localized value in shaping the net effect of a college degree.

Overall, these results indicate that both the anchoring effect of a degree and the catalyst effect of studying elsewhere are most pronounced for individuals originating from less economically advantaged areas, where the relative gains from migration or the relative boost provided by a local degree are largest.

6.5 Heterogeneity by Gender

Gender differences in migration are prominent in China, particularly among the non-college-educated, where men have historically exhibited higher rates of labor migration due to cultural norms and traditional household roles. This leads to two distinct hypotheses. First, given that non-college-educated men have a higher baseline propensity for temporary labor migration, we expect the negative, “anchoring” effect of a college degree *per se* to be more pronounced for them. In contrast, we hypothesize that the positive effect of the out-of-province college experience will be gender-neutral, as the mechanisms of location-specific learning and attachment formation are less likely to vary systematically by gender.

We test if our effects vary by gender by interacting our treatment variables (*College* and *CollegeOut*) with a male indicator. The results in column 4 show that the negative, “anchoring” effect of a college degree is significantly stronger for men ($College \times \mathbb{I}[\text{Male}]$ is negative). However, the effect of out-of-province enrollment does not differ by gender ($CollegeOut \times \mathbb{I}[\text{Male}]$ is insignificant). This suggests that while acquiring general human capital affects men’s and women’s migration decisions differently, the experience of living in a new location during college shapes subsequent mobility for both genders in a similar way.

6.6 Discussion

Our analysis reveals distinct migration patterns driven by skill level and educational experience in China. Lower-skilled young adults exhibit high geographic mobility, consistent with a strategy of temporary migration: their mobility is less deterred by *hukou* stringency or housing costs in destination cities because they minimize local expenditures through temporary housing and by remitting income home, thereby prioritizing immediate wage gains. In contrast, the migration of college graduates is strongly shaped by enrollment location, which functions as an “experience good.” By allowing students to learn about a location’s intangible benefits—career prospects, quality of life, social networks—the residential experience reduces the uncertainty associated with a permanent move, often encouraging settlement after graduation.

These findings have significant implications for both theory and policy. Theoretically, our result that a college degree can act as a “local anchor” provides a novel micro-foundation for

recent spatial equilibrium models that help explain why strong local ties can lead to persistent regional income disparities (Zabek, 2024). This underscores that education policy, within certain institutional and social contexts, can be a powerful, though often overlooked, driver of the local attachments that shape the economic landscape.

From a policy perspective, our results shed light on a classic efficiency-equity trade-off (Kennan, 2015). Policies that prioritize equity by expanding local college access might inadvertently strengthen the “local anchor” effect, potentially trapping talent in lower-opportunity regions. Conversely, policies that implicitly favor efficiency by concentrating educational resources or encouraging mobility (leveraging the “location experience” effect) risk exacerbating brain drain and regional inequality (Huang, Xing, and Cui, 2022; Fabre, 2023; Fajgelbaum and Gaubert, 2020; Yang, 2021). Effective interventions must also look beyond simple fiscal metrics and account for both the human capital and the location-specific experience effects that our paper isolates, while also considering crucial non-monetary factors, such as social norms and policy frictions.

Finally, our estimates should be interpreted in the context of our research design. The findings are LATE for the marginal students who entered college during China’s mass expansion. The negative “local anchoring” effect is likely most relevant for developing economies where institutional frictions are high; it may not generalize to developed countries with higher baseline mobility or to expansions of elite universities, where a degree is a more nationally portable signal. We do not interpret our finding as a universal law but as a powerful demonstration of how the returns to human capital are fundamentally shaped by the institutional context in which they are deployed.

7 Conclusion

Leveraging China’s late-1990s college expansion as a natural experiment, this study disentangles the distinct effects of acquiring a college degree (general human capital) from the experience of enrolling out-of-province (location-specific experience) on youth migration. Contrary to conventional wisdom, we find that while the location experience is a powerful catalyst for migration, the degree itself acts as a “local anchor,” slightly reducing migration propensity. This paradox arises because non-college graduates exhibit higher baseline mobility, often engaging in temporary migration shaped by institutional frictions like the *hukou* system, whereas the location

experience functions as an “experience good” for graduates, revealing a place’s intangible benefits and lowering the uncertainty of a permanent move.

These findings highlight the complex interplay between human capital, location experience, and institutional context in shaping the spatial allocation of talent. For policymakers, they underscore a fundamental trade-off. While local university investments may anchor talent, the geographic distribution of educational opportunities is a key determinant of efficient sorting and regional inequality. Effective policies must therefore look beyond simple fiscal and human capital metrics to account for the powerful role of location-specific experience.

Our analysis identifies partial equilibrium effects during early careers, pointing to two key avenues for future research. First, examining how these effects evolve over the life cycle, particularly in response to ongoing institutional changes like *hukou* reforms, is crucial. Second, a general equilibrium analysis accounting for the expansion’s aggregate impacts on wages, housing prices, and migration patterns would provide a more complete picture.

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Appendix

A.1 Sample Construction

We construct the analytical sample from the 2010 Chinese population census microdata via sequential filters. Starting with 4,636,871 individuals, we restrict the sample to China-born individuals, dropping 25 foreign-born observations and yielding 4,636,446 individuals. Because our analysis centers on the higher-education expansion, we retain only those with at least a high school education, removing 3,549,661 observations and leaving 1,086,785 individuals.

We next focus on cohorts for whom educational and residential histories can be reliably traced—those who would have been college-age during our study period (college cohorts 2003–2005). This step removes an additional 987,475 observations, resulting in 99,310 individuals. We then exclude 10,309 individuals who had not completed their education by the census date, leaving 89,001 observations.

Finally, we infer birth cities and college enrollment locations using *hukou* registration patterns and residential histories. This procedure excludes 19,335 individuals whose pre-college residence cannot be reliably identified, resulting in a sample of 69,666 observations. An additional 6,168 observations are excluded from the regression analysis due to missing city-level variables. This yields a final analytical sample of 63,498 individuals, which represents 18.9% of the initial eligible population aged 23–26 in the census.

A.2 Birth City Inference

Our analysis links individuals to attributes of their birth city and province. While the census reports birth province—allowing a direct match to provincial characteristics—it does not record birth city. We therefore infer birth city using three census fields: birthplace category, current county of residence, and the city of *hukou* registration.

The census classifies birthplace as: (1) the current county of residence; (2) a different county within the current province (county code unknown); or (3) a province other than the current one of residence (province code known). We apply the following rules to infer the birth city information:

1. **Born and currently residing in the same county:** The birth city is the individual's

current city of residence, which also matches the *hukou* city.

2. Born in a different county within the current province of residence:

- (a) If an individual's *hukou* city differs from their current city of residence, we assume their *hukou* city is their birth city. This applies to 60,622 individuals who likely moved within their province without updating their *hukou*. It is important to note that these individuals are not considered cross-province migrants.
- (b) We excluded 8,133 individuals who had registered their *hukou* in their current city of residence. For these individuals, we could not determine their birth city as their *hukou* registration might be updated post-migration and no longer indicates their origin.

3. Born in a different province from the current province of residence:

- (a) If an individual's *hukou* registration province matches their birth province, we assume their birth city is the city of their *hukou* registration.¹⁵
- (b) We exclude 1,601 individuals whose *hukou* province matches their current residence province rather than their birth province, as this prevents reliable identification of their birthplace city. These individuals have likely relocated and updated their *hukou* records to reflect their current residence.
- (c) We exclude 9,601 individuals who moved after college and were born in a province other than their current province of *hukou* registration.

A potential concern is that excluding individuals during our birth city inference introduces selection bias. We assess this by conducting a balance test, regressing key observable characteristics on an indicator for being dropped from the sample. The results in Table A.1 show no systematic differences between our analytical sample and the excluded individuals. The coefficients on key characteristics are statistically insignificant and economically small, providing confidence that our sample construction methodology does not introduce significant bias.¹⁶

¹⁵While some individuals may originate from a different city within their birth province than their *hukou* registration city, this is uncommon among young adults.

¹⁶While the age difference is marginally statistically significant (0.007 years), its magnitude (less than 3 days) is economically negligible.

Table A.1—Balance Test

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Han ethnicity	Male	Age	Population	Urban employment	GDP growth	Employment growth
Dropped	0.012 (0.009)	-0.006 (0.006)	0.007* (0.003)	-1.304 (2.877)	-73.071 (77.625)	-0.002 (0.003)	-0.005 (0.004)
Observations	89,001	89,001	89,001	89,001	89,001	89,001	89,001

Note: This table assesses potential selection bias arising from our birth city inference procedure. It compares observable characteristics ($\mathbf{X}$) between the final analytical sample ($N=63,498$) and individuals excluded because their birth city could not be inferred ($N=25,503$). Each column reports the coefficient δ from a separate OLS regression: $X = \alpha + \delta \times \mathbb{I}(\text{Dropped}) + \epsilon$, estimated on the combined sample ($N=89,001$), where $\mathbb{I}(\text{Dropped}) = 1$ for excluded individuals. Standard errors, clustered at the birth-province level, are in parentheses.

A.3 Additional Checks

We provide direct empirical support for the validity of our identification strategy and results through several checks.

First, an event-study analysis, presented in Figure A.1, compares key provincial economic outcomes between above- versus below-median enrollment expansion provinces before and after the 1999 policy shock. The absence of statistically significant differences between the two groups in the pre-policy period (1995-1998) supports the parallel trends assumption underlying our design.

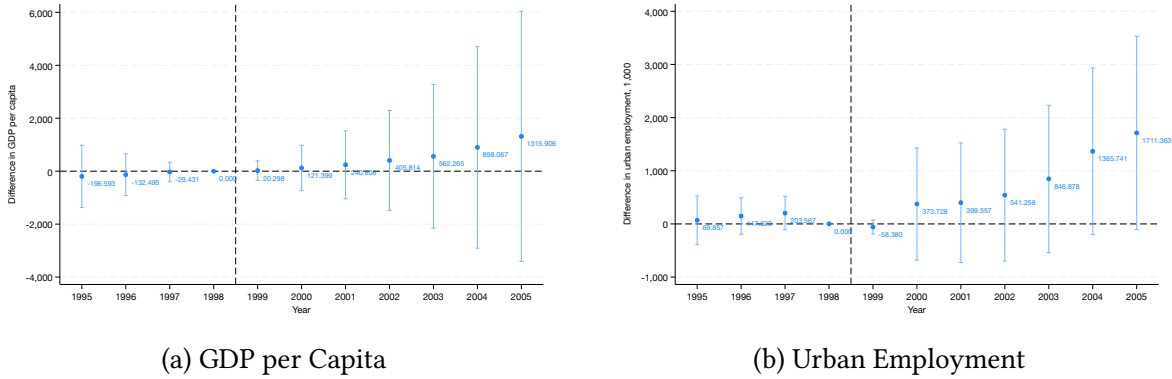
Second, we test whether our instruments (IR and OR) are correlated with other time-varying provincial characteristics that could potentially confound our estimates. Table A.2 presents regressions of the instruments on provincial population, GDP growth, and employment growth. We find no systematic relationship, providing further reassurance that our instruments capture exogenous variation specific to the college expansion.

Finally, Figure A.2 presents results from a leave-one-out analysis. We re-estimate our main 2SLS model 28 times, each time excluding one birth province. The resulting coefficient estimates for both College and CollegeOut cluster tightly around the full-sample benchmarks, demonstrating that our main findings are not driven by any single province.

A.4 The Expected Skill Premium Gain (Δw_j)

To construct our measure of the potential skill-premium gain from migration, we use individual-level monthly earnings data from the 2005 population census. From this source, we calculate

Figure A.1—Pre-Trend Analysis: Provincial Characteristics



Notes: This figure presents event-study plots comparing provinces with above- vs. below-median enrollment expansion (1995-2005). The vertical line indicates the 1999 policy implementation. The absence of statistically significant differences between the two groups in the pre-policy period (1995-1998) supports the parallel trends assumption.

Table A.2—Provincial Characteristics and College Expansion Shocks

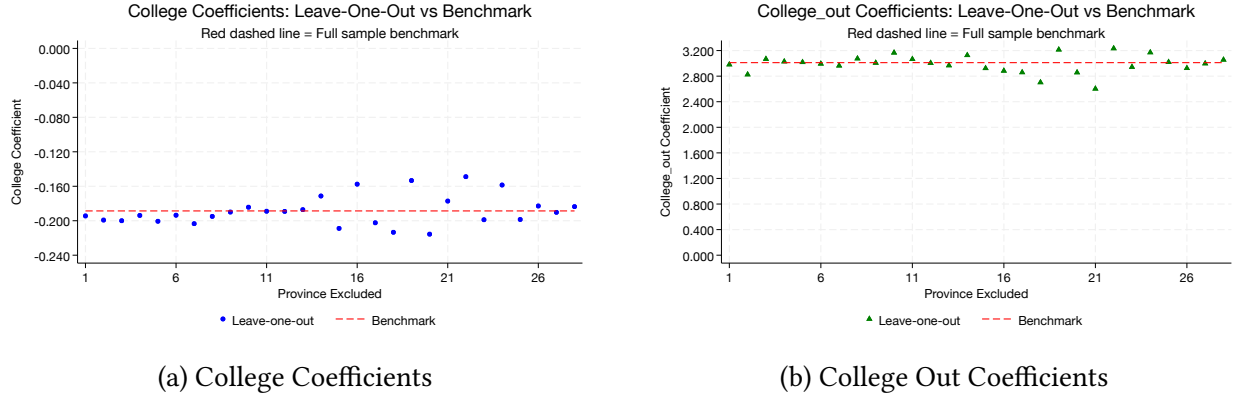
	(1) <i>IR</i>	(2) <i>OR</i>	(3) <i>IR</i>	(4) <i>OR</i>
Population (million)	0.012 (0.009)	0.002 (0.003)	0.014 (0.009)	0.004* (0.003)
Provincial GDP growth rate	-0.049 (0.116)	-0.083** (0.036)	-0.097 (0.193)	-0.055 (0.089)
Provincial employment growth rate	-0.039 (0.083)	0.003 (0.014)	-0.096 (0.085)	-0.017 (0.015)
Province FE	✓	✓	✓	✓
Year FE			✓	✓
Observations	155	155	155	155
R^2	0.879	0.958	0.897	0.967

Note: This table presents OLS estimates of the relationship between provincial characteristics and our constructed college expansion shocks. The dependent variables are the predicted in-province enrollment rate (*IR*) and out-of-province enrollment rate (*OR*) from equations (3) and (4). The sample covers 31 provinces from 2001 to 2005. Robust standard errors are in parentheses.

province-level average monthly wages for college-educated and non-college-educated workers using employed, prime-age (16–60) individuals who graduated after 2005 and reported positive earnings. These 2005 wage levels serve as a proxy for the expected earnings faced by cohorts making career and location decisions shortly thereafter.¹⁷ The skill premium in each province j is

¹⁷In 2005, higher wages were predominantly concentrated in major cities such as Beijing, Shanghai, and Guangzhou, as well as in coastal provinces such as Jiangsu, Zhejiang, and Fujian. In contrast, inland provinces and northwestern

Figure A.2—Leave-One-Out Analysis: Coefficient Stability Across Provinces



Notes: This figure shows the sensitivity of coefficient estimates to the exclusion of individual provinces. The red dashed line represents the benchmark coefficient from the full sample regression. Each point shows the coefficient estimate when the corresponding province is excluded.

defined as the relative wage difference between college-educated (w_j^H) and non-college-educated (w_j^L) workers in province j : $w_j = (w_j^H - w_j^L)/w_j^L$.

To calculate the expected skill premium in potential destinations, we use historical migration patterns from the census to construct weights. Let M_{jk} be the number of migrants from province j residing in province k . The probability of migrating from j to k , conditional on migrating, is the share M_{jk}/M_j , where $M_j = \sum_{k=1, j \neq k}^{31} M_{jk}$. The expected skill premium gain from migration for an individual born in province j , Δw_j , is the difference between the weighted-average skill premium across all potential destinations and the skill premium in their home province:

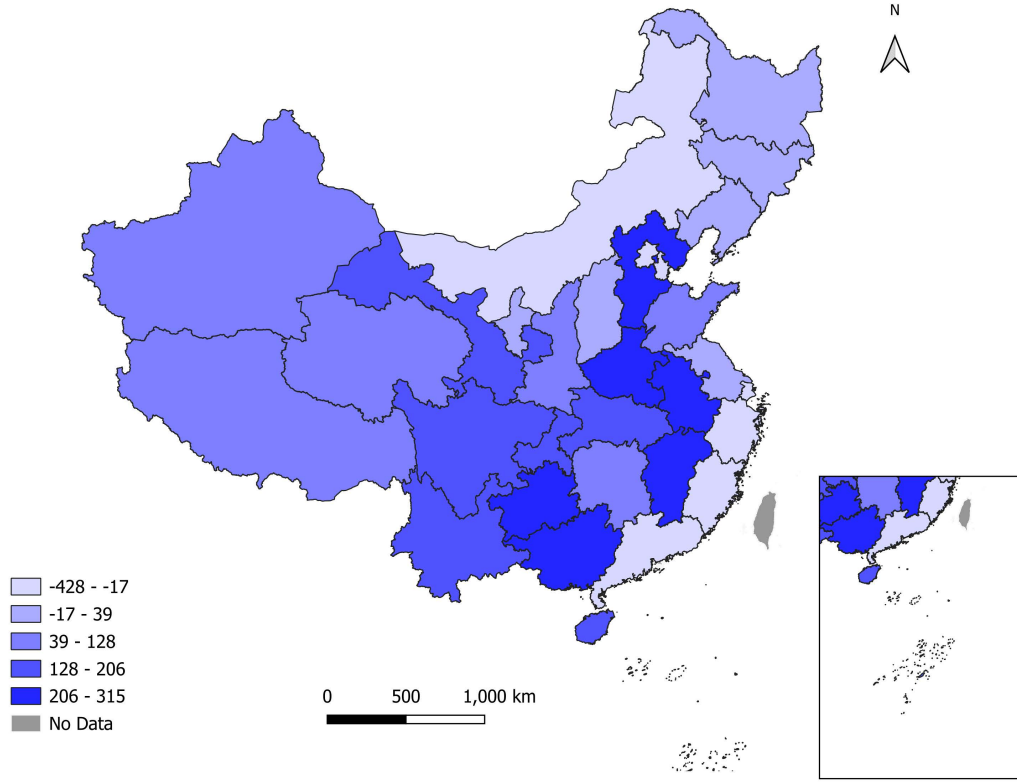
$$\Delta w_j = \sum_{k \neq j} w_k \times \frac{M_{jk}}{M_j} - w_j \quad (\text{A.1})$$

Figures A.3 and A.4 map the expected wage differentials for non-college (Δw_j^L) and college-educated workers (Δw_j^H), respectively. A positive value indicates that potential destination provinces offer, on average, a higher wage than an individual's home province.

The maps reveal substantial geographic heterogeneity. For non-college workers, the largest potential wage gains from migration are concentrated in western and central China (excluding resource-rich provinces like Shanxi and Inner Mongolia), while workers from economic centers,

regions exhibited lower wage levels for non-college-educated individuals. Notably, provinces with substantial ethnic populations tended to offer higher wages to non-college-educated individuals, possibly due to favorable policies and a thriving tourism industry.

Figure A.3—Expected Migration Wage Differential for Non-College-Educated Workers (Δw_j^L)

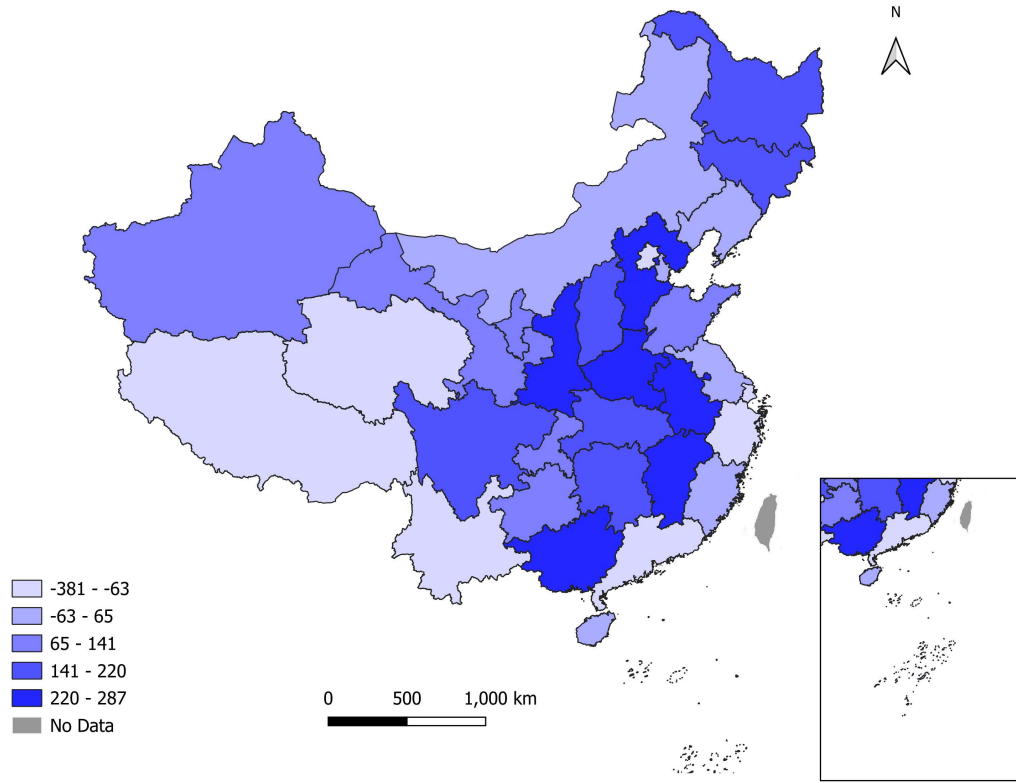


Notes: This map shows the expected monthly earnings differential, Δw_j^L , in yuan for a non-college-educated individual from province j considering migration. This differential is the difference between an expected migration wage—the average wage for non-college-educated workers in all possible destination provinces, weighted by historical non-college-educated migration shares from province j to each destination—and the wage for non-college-educated workers in the home province (w_j^L). The formula is $\Delta w_j^L = \sum_{k \neq j} w_k^L \times \frac{M_{jk}^L}{M_j^L} - w_j^L$, where the weight, $\frac{M_{jk}^L}{M_j^L}$, is the historical share of non-college-educated migrants from province j who reside in province k . The term M_{jk}^L is the number of non-college-educated migrants from province j in a specific destination k , and M_j^L is the total number of non-college-educated migrants from province j . All wage and migration data are from the 2005 population census.

such as Beijing, Shanghai, and Guangdong, face minimal expected gains or potential losses.

A distinct pattern emerges for college-educated workers. For this group, high potential wage gains are concentrated more narrowly in central China. Notably, potential wage gains in western provinces are smaller for college graduates than for their non-college-educated peers, potentially reflecting local policies aimed at retaining skilled talent in those regions.

Figure A.4—Expected Migration Wage Differential for College-Educated Workers (Δw_j^H)



Notes: This map shows the expected monthly earnings differential, Δw_j^H , in yuan for a college-educated individual from province j considering migration. This differential is the difference between an expected migration wage—the average wage for college-educated workers in all possible destination provinces, weighted by historical college-educated migration shares from province j to each destination—and the wage for college-educated workers in the home province (w_j^H). The formula is $\Delta w_j^H = \sum_{k \neq j} w_k^H \times \frac{M_{jk}^H}{M_j^H} - w_j^H$, where the weight, $\frac{M_{jk}^H}{M_j^H}$, is the historical share of college-educated migrants from province j who reside in province k . The term M_{jk}^H is the number of college-educated migrants from province j in a specific destination k , and M_j^H is the total number of college-educated migrants from province j . All wage and migration data are from the 2005 population census.

A.5 The Expected Housing Price Differential (Δhp_j)

To assess regional disparities in housing costs, we collected data on the average annual sale prices of commodity housing across 36 major Chinese cities, including all 4 direct-administered municipalities (Beijing, Tianjin, Shanghai, and Chongqing) and 27 provincial capitals.¹⁸ The data, obtained from the National Bureau of Statistics, covers the period from 2002 to 2005. This

¹⁸In China, commodity housing refers to housing sold at market price, as opposed to subsidized units with resale restrictions.

timeframe precedes the graduation of our sample cohorts, ensuring that our measure captures *ex-ante* housing cost expectations.

Following a method analogous to the wage gain calculation, we use historical migration shares (M_{jk}/M_j), the proportion of migrants from each origin province j to destination province k , as weights. The potential housing cost change from migration (Δhp_j) for individuals born in province j is the difference between the weighted average *ex-ante* housing prices in all potential destination provinces (hp_k) and the average *ex-ante* housing price in the home province (hp_j):

$$\Delta hp_j = \sum_{k \neq j} hp_k \times \frac{M_{jk}}{M_j} - hp_j \quad (\text{A.2})$$

where a positive Δhp_j indicates that housing is, on average, more expensive outside an individual's home province. Figure A.5 maps this measure, showing that individuals from inland provinces, such as Guangxi, Hunan, and Jiangxi, face the largest potential cost increases from migrating, while those from developed regions, such as Beijing, Shanghai, Guangdong, and other coastal provinces, face potential cost savings.

A.6 The Expected *Hukou* Stringency Differential ($\Delta hukou_{jt}$)

To measure provincial differences in the stringency of mobility restrictions, we leverage the *hukou* reform index developed by Fan (2019). This index assigns a score (0-6) to each city based on local policy documents, with higher scores denoting more lenient policies (i.e., easier access to a local *hukou*). We first aggregate this city-level index ($hukou_c$) to the provincial level (R_j) by taking a weighted average:

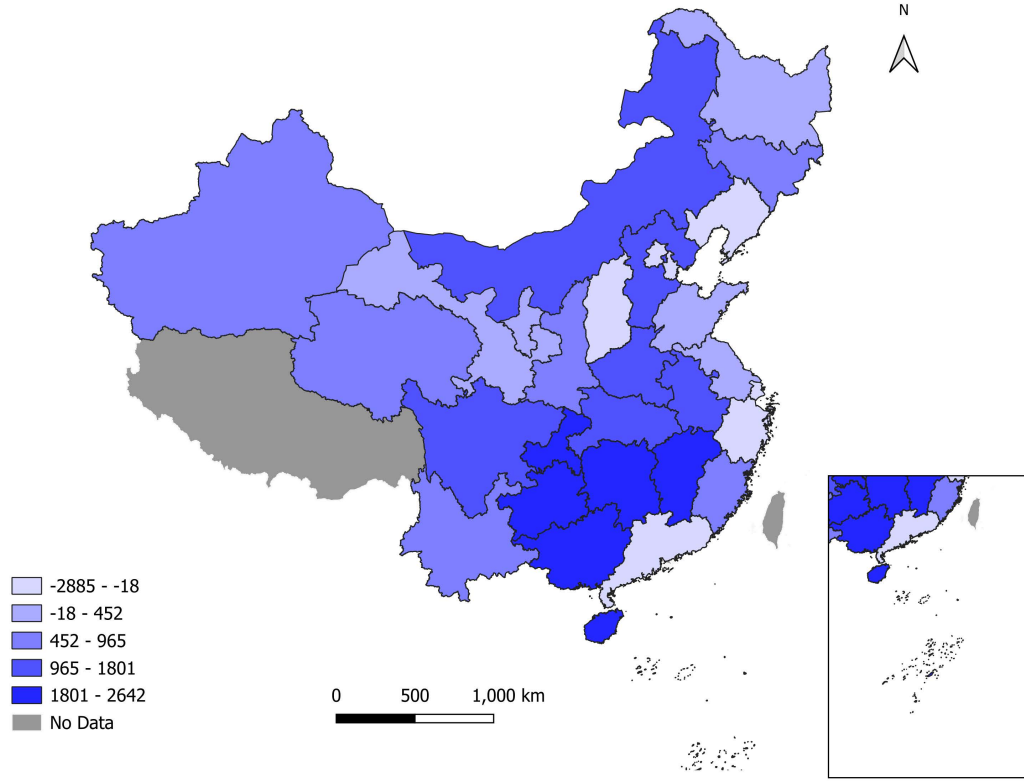
$$R_j = \sum_c hukou_c \times \omega_{cj}$$

where the weights are based on each city's share of the total migrant population within its province (using the 2000 census):

$$\omega_{cj} = \frac{\text{Total number of migrants in city } c}{\text{Total number of migrants in province } j}$$

Following the same logic as expected wage and housing price differentials upon migration, we

Figure A.5—Spatial Distribution of Expected Housing Price Differentials (Δhp_j), 2002-2005



Notes: This map shows the expected housing price differential (Δhp_j) from migration in yuan per square meter (yuan/m^2) for an individual from province j . The differential is calculated as: $\Delta hp_j = \sum_{k \neq j} hp_k \times \frac{M_{jk}}{M_j} - hp_j$ where hp_j is the average housing price in an individual's home province. The expected destination price is a weighted average of housing prices in all other potential destination provinces (hp_k), with weights determined by the historical share of migrants from province j that reside in each destination province k (M_{jk}/M_j). Housing price data are from the National Bureau of Statistics, and migration shares are calculated using the 2005 population census. Tibet is excluded due to data unavailability.

then calculate the expected difference in *hukou* stringency upon migration in year t , $\Delta hukou_{jt}$:

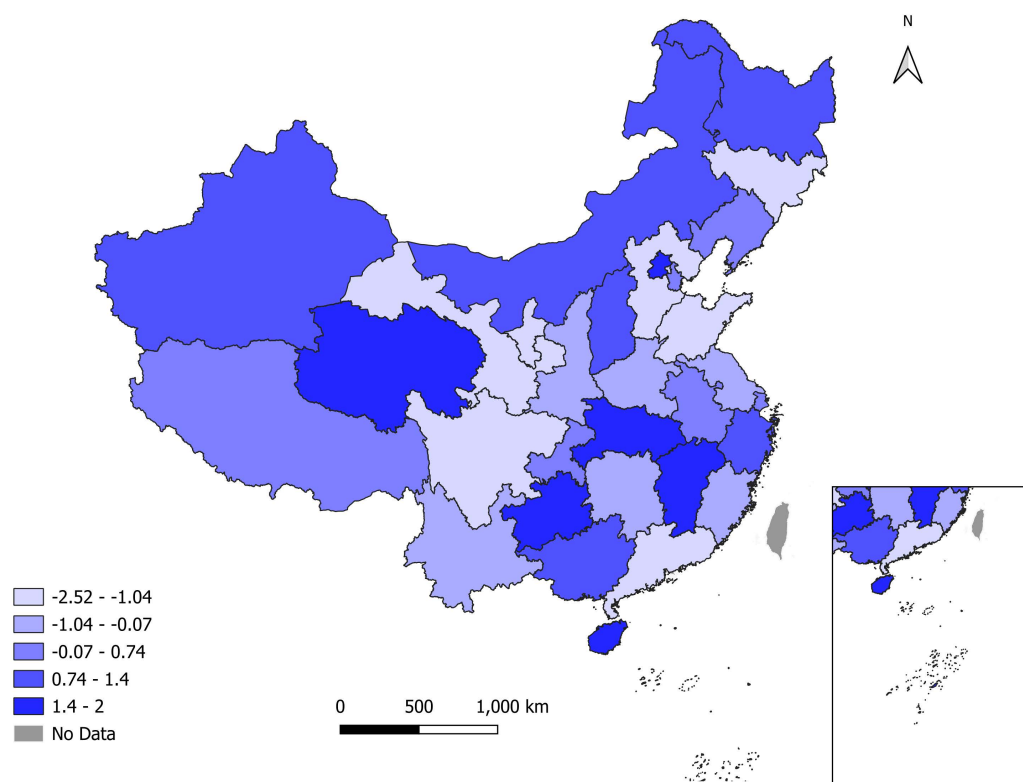
$$\Delta hukou_{jt} = \sum_{k \neq j} R_{kt} \times \frac{M_{jk}}{M_j} - R_{jt} \quad (\text{A.3})$$

where a larger value of $\Delta hukou_{jt}$ indicates that potential destination provinces, on average, impose less stringent *hukou* restrictions than an individual's birth province j .

Figure A.6 maps the spatial distribution of the expected difference in *hukou* stringency ($\Delta hukou_{jt}$) for an individual migrating from province j , where darker shades indicate a gain in

leniency. Positive values are concentrated in coastal megacities and advanced economic regions, such as Beijing, Shanghai, and the Yangtze and Pearl River Deltas. Individuals originating from these areas face the strictest baseline *hukou* regimes; consequently, migration typically entails moving to jurisdictions with more relaxed settlement policies. In contrast, interior and western provinces exhibit negative values, indicating that migrants from these relatively open regions typically move to destinations with stricter entry barriers. The Northeast and parts of central China show values near zero, suggesting that the expected *hukou* stringency in potential destinations is comparable to that of the home province.

Figure A.6—Expected Difference in *Hukou* Stringency upon Migration ($\Delta hukou_{jt}$)



Notes: This figure maps the spatial distribution of the expected difference in *hukou* stringency ($\Delta hukou_{jt}$) for an individual migrating from province j . The variable is constructed using the *hukou* reform index from [Fan \(2019\)](#), which scores cities on a scale of 0 to 6, with higher scores indicating more lenient settlement policies. Provincial-level stringency is aggregated from city-level indices weighted by the city's share of the province's total migrant population in the 2000 census. A higher (positive) value of $\Delta hukou_{jt}$ indicates that potential destination provinces have, on average, more lenient *hukou* policies than the individual's birth province.